MIL-F-14052C
20 March 1970
SUPERSEDING
MIL-F-14052B
7 January 1966

MILITARY SPECIFICATION

FIBER ROPE ASSEMBLY, SINGLE LEG

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers requirements for single leg, fiber rope assemblies.

* 1.2 Classification.- The fiber rope assemblies shall be of the following types and sizes as specified (see 6.2).

Type I - Spliced loops on each end.
Size A - 20 feet long
Size B - 34 feet long

Type II - One end with spliced loop other rope end served.
Size A - 15 feet long

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

T-R-605	-	Rope, Manila and Sisal.
T-T-871	-	Twine, Cotton, Wrapping.
CCC-C-429	-	Cloth, Osnaburg, Cotton.
CCC-C-467	-	Cloth, Burlap, Jute (or Kenaf).
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible.

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MIL-L-1709	-	Line, Tent.
MIL-T-40625	-	Tubing, Bias Sewn (Burlap or Osnaburg) Cloth.

FSC 4020

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials.-

* 3.1.1 Rope.- The rope used in the fabrication of the assemblies shall be three strand manila, conforming to type M, class 2 of T-R-605. The size of rope for type I, size A and B assemblies shall be 3-1/2 inch circumference rope. The size of rope for type II, size A assembly shall be a 1-1/2 inch circumference rope.

* 3.2 Design.-

* 3.2.1 Type I.- Type I assemblies shall consist of one continuous length of manila rope specified in 3.1.1 with eye spliced loops (see 3.3.3.1) on each end as shown in figure 1.

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* 3.2.2 Type II.- Type II assemblies shall consist of one continuous length of manila rope specified in 3.1.1, with one rope end with an eye spliced loop (see 3.3.3.1) and the opposite rope end served (see 3.3.3.2).

3.3 Construction.-

* 3.3.1 Length.- The length of the assemblies shall be as specified in table I. The assemblies shall be in continuous lengths without knots or splices or otherwise secured to make a continuous length. The length for type I assemblies shall be determined by measuring from the farthest extremity of each eye spliced loop (see "L", figure 1). The length of type II assemblies shall be determined by measuring the farthest extremity of the eye splice to the opposite end of the assembly.

* 3.3.2 Loop size.- The inner perimeter of each eye splice loop shall be as specified in table I (see "P", figure 1).

TABLE I.- Length and loop size requirements

Type and size	Length (feet)	Loop size (inches)
Type I		
Size A	20 feet ± 1-1/2 inches	43 inches ± 1-1/2 inches
Size B	34 feet ± 1-1/2 inches	28 inches ± 1-1/2 inches
Type II		
Size A	15 feet ± 1-1/2 inches	8 inches ± 1-1/2 inches

3.3.3 End finish.-

3.3.3.1 Splicing.- The eye splice shall be formed by bending the end of the rope upon itself to form an eye and then spliced as follows to form a neat appearing eye splice. Unlay the rope a suitable distance and make an eye of required size; serve each strand and tuck the bottom strand under one strand against the lay, and tuck the middle strand under the next strand in the same manner. Turn the splice over and tuck the remaining strand as specified above. Tuck each strand a second time against the lay, over once and under once, and repeat until four full strand tucks are in the splice. Taper the ends of the strands to approximately one-half size and tuck each strand. Trim the ends of the strand ends to 1/2 inch and roll under foot, or tap with a hammer to insure a neat appearing close-fitting splice. Splices shall be formed without bulges, loose tucks or untrimmed strand ends.

* 3.3.3.2 Serving (type II only).- One end of the rope of the type II assembly shall be served in conformance with figure 2 of MIL-L-1709, with 5 ply cotton twine conforming to type II, class 1 of T-T-871.

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* 3.4 Marking for identification.- A corrosion resistant .016 inch minimum thick steel tag shall be securely fastened to the assembly adjacent to one of the eye spliced loops by a corrosion resistant .047 inch minimum diameter steel wire. Marking on the tag shall be in accordance with MIL-STD-130 and shall contain the following information:

Fiber Rope Assembly, Single Leg
Federal Stock Number
Type and Size
Date of Manufacture
U. S. A.

3.5 Workmanship.- The fiber rope assemblies shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in the specification or applicable purchase documents. In addition, inspection shall be performed for the requirements in table II and a supplier's certificate of compliance thereto shall be furnished and will be accepted for the requirement.

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TABLE II.- Component test

Characteristic	Paragraph reference
Corrosion resistant tag	3.4
Corrosion resistant wire	3.4

4.2.2 Examination of the end item for visual defects.- The defects listed in table III shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. The sample unit for this examination shall be one complete fabricated assembly. The lot size for this examination shall be expressed in units of assemblies each. The acceptable quality level shall be 2.5 major defects and 6.5 total defects (major and minor combined) per 100 units. The inspection level shall be level II.

TABLE III.- Visual examination defects

Examine	Defect	Classification Major	Minor
Finished item	Not as specified	X	
Rope	Cut, any	X	
	Chafed or damaged	X	
	Kinks, broken or loose ends, bulged strands	X	
	Not of continuous length or contains knots, splices or otherwise secured to make a continuous length	X	
	Length less than or more than length specified (including permitted tolerances).	X	
Cleanness	Spot, stain 1/		X
Eye splice	Not accomplished as specified	X	
	Number of tucks less than specified	X	
	Tucks of splices not made against lay of the rope	X	
	End strands of splices not tapered or trimmed		X
	Loose tucks, bulged splices or uneven splicing	X	

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TABLE III.- Visual examination defects (cont'd)

Examine	Defect	Classification Major	Minor
Served end (when applicable)	Not secured or not accomplished as specified		X
	Not served	X	
	Number of turns less than specified		X
	Loose ends not tucked in lay of rope		X
Identification	Omitted, incorrect, illegible		X
	Insecurely attached		X
	Not as specified		X

1/ Clearly visible at normal inspection distance (approximately 3 feet).

4.2.3 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing and marking requirements complies with Section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping bundle fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping bundles fully prepared for delivery. The lot size shall be the number of bundles in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5 defects per 100 units.

Examine	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged.
Workmanship	Inadequate application of components, such as: inadequate barrier material wrap and cloth overwrap. Improper wire tying and twine sewing. Quantity per bundle more or less than required.
Packaging	Not as specified.
Packing	Not as specified.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.3).

5.1.1 Level A.- Each rope assembly shall be folded in half and the loops tied with twine conforming to type II, 24 ply of T-T-871. Six folded and tied assemblies shall be placed together and tied at the middle. The bundle shall again be folded in half, and the free ends tied together.

5.1.2 Level C.- Rope assemblies shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.3).

5.2.1 Level A.- Six rope assemblies, of one size only packaged as specified in 5.1, shall be wrapped with waterproof barrier material conforming to B3(a) of PPP-B-1055, with a minimum overlap of 6 inches at the joints. The assemblies shall then be placed in bias-sewn tubing or a bag (presewn one end) made from cotton osnaburg cloth conforming to class 2 of CCC-C-429 or jute (or kenaf) burlap cloth conforming to class 3 of CCC-C-467. The bias-sewn tubing shall conform to MIL-T-40625. The cloth covered bundles shall have the open end closed with two double-looped wire ties. The wire ties shall be not less than 6 inches long, 0.072-inch thick galvanized soft iron or steel wire with a 1/2-inch diameter formed eye at each end. The first wire tie shall be applied as close to the bundle as possible. The second wire tie shall be applied at a distance approximately 1 inch from the first wire tie. The twisted ends of the wire ties shall alternate and face in opposite directions. Alternatively, the assemblies shall be overwrapped and sewn in the specified cotton osnaburg cloth or jute (or kenaf) burlap cloth. The wrapping shall be securely sewn with 16-ply cotton twine or 3-ply jute twine with approximately one stitch to the inch and every third stitch knotted.

5.2.2 Level B.- Six rope assemblies, of one size only packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the requirement for the PPP-B-1055 barrier material shall not apply.

5.2.3 Level C.- Rope assemblies, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Bundles shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.- The fiber rope assembly is intended for cargo and material handling operations.

6.2 Federal Stock Numbers.- The federal stock number of these items are:

Type I, Size A - FSN 4020-270-8678
Type I, Size B - FSN 4020-368-2936
Type II, Size A - FSN 4020-298-3394

6.3 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type and size (see 1.2 and table I).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. Whether special marking is required (see 5.3).

* 6.4 Marginal identification.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SH
Air Force - 82

Preparing activity:

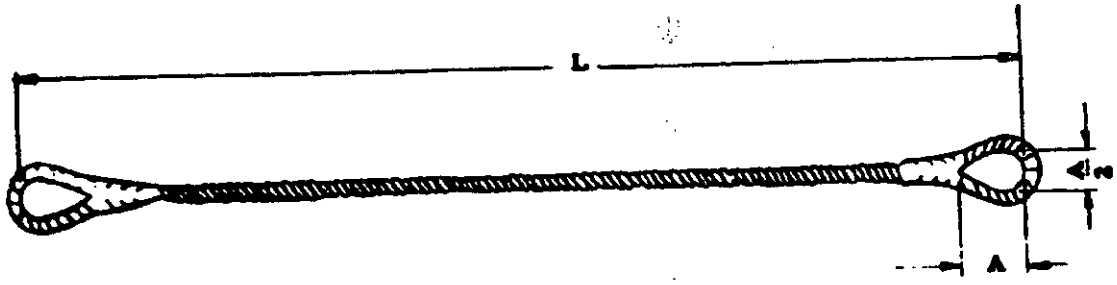
Army - GL

Project No. 4020-0131

Review Activities:

Army - ME
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L = Length of assembly.
A and A/2 = Position of loop for measuring.
P = Loop size.



FIGURE 1--Measurement of assemblies.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-F-14052C Fiber Rope Assembly, Single Leg

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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